ALTER in MySQL ;

ALTER is one of the most important SQL commands because real databases **change over time**.

You create a table today. Tomorrow business asks:

- Add new column
- Rename old column
- Change datatype
- Remove column
- Add default values
- Add constraints
- Rename table

That is where **ALTER** is used.

GoodNotes / Interview Definition

ALTER is a DDL (Data Definition Language) command used to modify the structure of an existing database object such as a table, columns, constraints, or indexes.

Visual Meaning

CREATE TABLE = Build new house 🏠

ALTER TABLE = Renovate existing house 🔨

- add room

- remove room

- change room size

- rename room

Basic Syntax

ALTER TABLE table_name

action;

Internal Working (Important)

When MySQL runs ALTER:

Existing Table

↓

Reads metadata

↓

Changes structure

↓

May rebuild table (sometimes)

↓

Updates schema dictionary

↓

Done

⚠️ Some ALTER operations are fast metadata-only.
⚠️ Some require full table rebuild (slow on big tables).

Sample Table

```
CREATE TABLE employees (
  id INT,
  name VARCHAR(30),
  salary INT
);
```

---

1. ADDING COLUMNS

Syntax

```
ALTER TABLE employees
ADD column_name datatype;
```

Example

```
ALTER TABLE employees
ADD age INT;
```

Result

id | name | salary | age

Add Multiple Columns

```
ALTER TABLE employees
ADD city VARCHAR(20),
ADD doj DATE;
```

Add at Specific Position

```
ALTER TABLE employees
ADD age INT AFTER name;
```

id | name | age | salary

---

2. CHANGING DATATYPES

Used when old datatype is too small / wrong.

Syntax (MODIFY)

```
ALTER TABLE employees
MODIFY column_name new_datatype;
```

Example

```
ALTER TABLE employees
MODIFY salary DECIMAL(10,2);
```

Now salary can store:

55000.75

Visual Flow

salary INT

↓ ALTER MODIFY

salary DECIMAL(10,2)

---

3. RENAMING COLUMN

Modern MySQL

```
ALTER TABLE employees
RENAME COLUMN name TO full_name;
```

Result

id | full_name | salary

---

4. CHANGING COLUMN NAME + DATATYPE TOGETHER

Use CHANGE

ALTER TABLE employees

CHANGE name full_name VARCHAR(50);

Meaning:

old_name new_name datatype

name full_name VARCHAR(50)

---

5. RENAMING TABLE

ALTER TABLE employees

RENAME TO staff;

Now table name becomes staff.

Visual

employees table

↓

ALTER RENAME

↓

staff table

---

6. MODIFYING VALUES (Important Clarification)

ALTER changes **structure**, not row data.

To modify actual values use:

UPDATE employees

SET salary = 50000

WHERE id = 1;

Wrong Use

ALTER TABLE employees ...

❌ cannot update row values directly.

Visual Difference

ALTER = change table design

UPDATE = change row data

---

7. INSERTING DATA TYPES (Clarification)

Datatype is not inserted.
You assign datatype when creating or adding column.

Example:

ALTER TABLE employees

ADD bonus DECIMAL(8,2);

You are adding a **column with datatype**.

---

8. ADD DEFAULT VALUE

ALTER TABLE employees

ALTER COLUMN salary SET DEFAULT 10000;

(Depending on MySQL version, common way:)

ALTER TABLE employees

MODIFY salary INT DEFAULT 10000;

---

9. DROP COLUMN

ALTER TABLE employees

DROP COLUMN age;

Visual

Before: id | name | age | salary

After : id | name | salary

---

10. ADD PRIMARY KEY

ALTER TABLE employees

ADD PRIMARY KEY(id);

---

11. ADD UNIQUE KEY

```
ALTER TABLE employees
ADD UNIQUE(name);
```

---

12. ADD FOREIGN KEY

```
ALTER TABLE orders
ADD CONSTRAINT fk_emp
FOREIGN KEY(emp_id)
REFERENCES employees(id);
```

---

13. MAKE COLUMN NULL / NOT NULL

Allow NULL

```
ALTER TABLE employees
MODIFY salary INT NULL;
```

Force NOT NULL

```
ALTER TABLE employees
MODIFY salary INT NOT NULL;
```

---

14. CHANGE COLUMN POSITION

```
ALTER TABLE employees
MODIFY salary INT AFTER name;
```

Execution Thinking

Need new column? → ADD

Need remove column? → DROP

Need datatype change? → MODIFY

Need rename column? → RENAME COLUMN / CHANGE

Need rename table? → RENAME TO

Need row data change? → UPDATE

Real Company Use Cases

E-commerce

Need GST number:

ALTER TABLE customers ADD gst_no VARCHAR(20);

HR

Salary now needs decimals:

ALTER TABLE employees MODIFY salary DECIMAL(12,2);

Startup

Rename vague column:

name → full_name

Banking

Enforce unique account number.

Common Mistakes

1.

ALTER employees ...

❌ Must use ALTER TABLE

2.

Trying to change values with ALTER.

Use UPDATE

3.

Changing datatype causes truncation risk.

Example:

VARCHAR(100) → VARCHAR(5)

May cut data.

4.

Big tables may lock / take time.

Practice Problems

1

Add email VARCHAR(50) to students table.

2

Change salary from INT to DECIMAL(10,2)

3

Rename column name to full_name

4

Rename table emp to employees

5

Drop column age

Pro Tips

Use DESCRIBE before and after

DESC employees;

Backup before altering big tables.

On production tables:

ALTER can be risky if millions of rows.

Summary Table

Need	Command
Add column	ADD
Remove column	DROP
Change datatype	MODIFY
Rename column	RENAME COLUMN
Rename col + datatype	CHANGE
Rename table	RENAME TO
Change row values	UPDATE

Golden Rule

ALTER changes structure

UPDATE changes data

INSERT adds rows